

Product Requirements Document (PRD)

KisanGat

Smart Dairy Management & ERP System

Version: 1.0 (MVP)

Prepared By: Product Management Team

1. Product Overview

KisanGat is a web-based dairy management platform designed to help dairy owners digitize and automate their day-to-day operations. The platform provides a centralized system for managing milk collection from farmers, milk sales to customers, inventory tracking, payment settlements, billing, employee records, dairy product sales, and business analytics.

The primary goal of KisanGat is to replace manual registers, notebooks, and spreadsheets with a reliable digital solution that improves operational efficiency, transparency, and profitability.

2. Problem Statement

Most small and medium-sized dairies still operate using paper-based records or spreadsheets.

Common challenges include:

- Difficulty tracking daily milk collection.
- Manual calculation errors in farmer payments.
- Poor visibility into milk inventory.
- Inability to track customer dues effectively.
- Time-consuming bill generation.
- Lack of business analytics and reporting.
- No centralized record of dairy operations.

As the business grows, these challenges lead to revenue leakage, operational inefficiencies, and inaccurate financial reporting.

KisanGat aims to solve these problems through a unified dairy management platform.

3. Target Users

Primary User

Dairy Owner / Administrator

Responsible for:

- Managing milk collection
 - Tracking sales
 - Managing customer accounts
 - Processing farmer payments
 - Monitoring inventory
 - Generating reports
 - Managing employees
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Secondary Users

No secondary user roles will exist in Version 1.

The system will support only one authenticated Admin account.

4. Product Vision

To become the preferred digital operating system for dairy businesses by simplifying dairy operations, improving financial transparency, and enabling data-driven decision making.

5. Business Goals

Operational Goals

- Digitize dairy operations.
- Reduce manual calculations.
- Improve payment accuracy.
- Simplify billing processes.

Financial Goals

- Improve revenue tracking.
- Reduce outstanding dues.
- Increase inventory visibility.

User Goals

- Save administrative time.
 - Improve record accuracy.
 - Generate reports instantly.
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6. Core Features

Feature 1: Admin Authentication

Description

Secure access to the system through a single Admin login.

Capabilities

- Login
 - Logout
 - Change Password
 - JWT Authentication
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Feature 2: Dashboard

Description

Provide a real-time overview of dairy operations.

Dashboard Metrics

- Today's Milk Collection
 - Today's Milk Sales
 - Today's Revenue
 - Outstanding Dues
 - Farmer Payments Pending
 - Current Inventory
 - Dairy Product Sales
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Feature 3: Farmer Management

Description

Manage all farmer records.

Features

- Add Farmer
- Edit Farmer
- View Farmer Profile
- Farmer Ledger
- Farmer Collection History

Data

- Name
 - Mobile Number
 - Address
 - Bank Details
 - UPI Details
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Feature 4: Milk Collection Management

Description

Record milk supplied by farmers.

Features

- Daily Collection Entry
- Cow Milk Entry
- Buffalo Milk Entry
- Fat Percentage Recording
- Automatic Amount Calculation

Data Captured

- Date
 - Farmer
 - Milk Type
 - Quantity
 - Fat Percentage
 - Rate
 - Amount
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Feature 5: Customer Management

Description

Manage customers who purchase milk.

Features

- Add Customer

- Customer Profile
- Customer Ledger
- Purchase History

Customer Types

- Retail
 - Wholesale
 - Hotel
 - Shop
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Feature 6: Milk Sales Management

Description

Track milk sold to customers.

Features

- Daily Sales Entry
 - Customer-wise Sales
 - Cow Milk Sales
 - Buffalo Milk Sales
 - Mixed Milk Sales
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Feature 7: Inventory Management

Description

Track milk inventory automatically.

Inventory Logic

Opening Stock

+ Milk Collected

- Milk Sold = Remaining Stock

Features

- Daily Stock Tracking
 - Remaining Stock Calculation
 - Carry Forward Stock
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Feature 8: Customer Due Management

Description

Track customers who purchase milk on credit.

Features

- Due Tracking
- Outstanding Amount Calculation
- Payment Recording
- Due History

Payment Status

- Paid
 - Partial
 - Due
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Feature 9: Farmer Payment Management

Description

Automate farmer payment calculations.

Features

- Daily Settlement
- Weekly Settlement
- Monthly Settlement
- Payment History

Calculations

Total Milk Quantity × Applicable Rate

Feature 10: Dairy Product Management

Description

Manage products produced from milk.

Products

- Curd
- Buttermilk
- Ghee
- Butter

- Paneer

Features

- Product Inventory
 - Product Sales
 - Product Revenue Tracking
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Feature 11: Employee Records

Description

Maintain employee records.

Features

- Employee Information
- Attendance Records
- Salary Records

Note: Employees do not have login access in Version 1.

Feature 12: Billing & Invoicing

Description

Generate bills and statements.

Farmer Bills

- Daily
- Weekly
- Monthly

Customer Bills

- Daily
- Weekly
- Monthly

Formats

- PDF
 - Printable View
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Feature 13: Reports & Analytics

Reports

- Daily Collection Report
- Farmer Payment Report
- Customer Due Report
- Revenue Report
- Product Sales Report
- Inventory Report
- Expense Report

Analytics

- Collection Trends
 - Revenue Trends
 - Sales Trends
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7. User Stories

Authentication

As an Admin, I want to log into the system securely, so that unauthorized users cannot access dairy data.

Farmer Management

As an Admin, I want to add and manage farmers, so that I can maintain accurate supplier records.

Milk Collection

As an Admin, I want to record daily milk collection, so that I can calculate farmer payments accurately.

Customer Management

As an Admin, I want to maintain customer records, so that I can track purchases and dues.

Inventory

As an Admin, I want inventory to update automatically, so that I always know available milk stock.

Due Management

As an Admin, I want to track unpaid customer balances, so that I can collect payments efficiently.

Farmer Payments

As an Admin, I want farmer payments to be calculated automatically, so that settlement errors are minimized.

Billing

As an Admin, I want to generate customer and farmer bills, so that transactions remain transparent and documented.

8. MVP Scope

The MVP should include only the essential features required for a dairy owner to operate the business digitally.

Included in MVP

- Admin Login
 - Dashboard
 - Farmer Management
 - Milk Collection
 - Customer Management
 - Milk Sales
 - Inventory Management
 - Customer Dues
 - Farmer Payments
 - Dairy Product Management
 - Employee Records
 - Billing
 - Reports
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9. Out of Scope (Version 1)

The following features should NOT be developed in the MVP.

Multi-User Roles

- Accountant Login
- Employee Login

- Permission Management

Mobile Applications

- Android App
- iOS App

Customer Portal

- Customer Self-Service Dashboard

Farmer Portal

- Farmer Self-Service Dashboard

Integrations

- WhatsApp Integration
- SMS Notifications
- UPI Payments
- Payment Gateway

Advanced Technology

- AI Demand Forecasting
- IoT Milk Collection Devices
- GPS Tracking

Enterprise Features

- Multi-Dairy Support
 - Franchise Management
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10. Success Metrics

Business Metrics

- 100% digital transaction recording.
- 90% reduction in manual calculations.
- Accurate farmer payment generation.
- Accurate inventory tracking.

Product Metrics

- Dashboard load time under 3 seconds.
- Bill generation under 5 seconds.
- Report generation under 10 seconds.

User Metrics

- Daily active usage by dairy owner.
 - Reduced bookkeeping effort.
 - Faster monthly settlement process.
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11. Risks & Assumptions

Assumptions

- Dairy operations are managed by a single administrator.
- Internet connectivity is available.
- Users have basic computer knowledge.

Risks

- Frequent changes in milk pricing.
 - Data entry mistakes by operators.
 - Growth in business may require multi-user support.
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12. Future Roadmap

Phase 2

- Multi-user roles
- WhatsApp bill sharing
- SMS reminders
- UPI payment tracking
- Excel export

Phase 3

- Android application
 - Farmer portal
 - Customer portal
 - AI analytics
 - Demand forecasting
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Product Vision Statement

KisanGat will serve as a complete digital operating platform for dairy businesses, enabling efficient milk collection, payment management, inventory control, customer billing, and business analytics through a simple and centralized administration system.